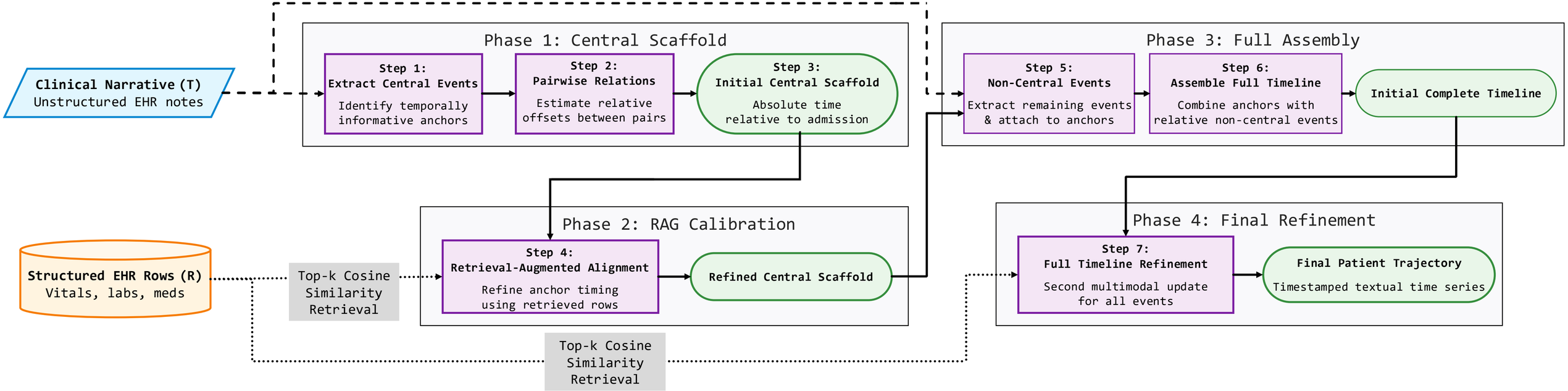
Clinical Narrative (T)
Unstructured EHR notes
Phase 1: Central Scaffold
Step 1:
Extract Central Events
Identify temporally informative anchors
Step 2:
Pairwise Relations
Estimate relative offsets between pairs
Step 3:
Initial Central Scaffold
Absolute time relative to admission
Phase 3: Full Assembly
Step 5:
Non-Central Events
Extract remaining events & attach to anchors
Step 6:
Assemble Full Timeline
Combine anchors with relative non-central events
Initial Complete Timeline
Structured EHR Rows (R)
Vitals, labs, meds
Top-k Cosine Similarity Retrieval
Phase 2: RAG Calibration
Step 4:
Retrieval-Augmented Alignment
Refine anchor timing using retrieved rows
Refined Central Scaffold
Phase 4: Final Refinement
Step 7:
Full Timeline Refinement
Second multimodal update for all events
Final Patient Trajectory
Timestamped textual time series
Top-k Cosine Similarity Retrieval